

PHISHING URL SECURITY AUDIT REPORT

CONFIDENTIAL

REF: PG-AUDIT-00349

Executive Cybersecurity Risk Assessment & Explainable AI (XAI) Diagnostic Audit

1. REPORT INFORMATION

Report Title:	Phishing URL Security Audit Report	Scan Reference ID:	#349 (PG-SCAN-00349)
Audit Date & Time:	August 20, 2026 at 15:39:45 UTC	Target Domain:	paypal-security-update.account-verify.xyz
Analyzed URL:	http://paypal-security-update.account-verify.xyz/signin.php?token=928103	ML Model:	Random Forest + XAI

2. FINAL VERDICT

■ PHISHING THREAT DETECTED

RISK LEVEL: CRITICAL

This URL exhibits malicious deception patterns characteristic of phishing, brand spoofing, or credential theft campaigns.

PREDICTION PHISHING	RISK LEVEL CRITICAL	CONFIDENCE 85.0%	PHISHING PROB. 85.0%	LEGITIMATE PROB. 15.0%
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3. URL SECURITY ANALYSIS

Security Characteristic	Observed Value / Diagnostic Status	Security Characteristic	Observed Value / Diagnostic Status
HTTPS Encryption	No (Insecure HTTP)	Domain Name	paypal-security-update.account-verify.xyz
URL Length	72 characters (Normal)	Domain Length	41 characters
Number of Subdomains	1 level(s)	Suspicious Keywords	Yes — 3 keyword(s) (signin, verify, account)
IP Address Usage	No (Standard DNS Host)	Suspicious Structure	Yes (Hyphenated Brand Spoofing)
TLD Abuse Rating	80% Risk (High-Risk Abuse TLD)	Shannon Entropy	4.74 bits (High Randomness / Obfuscated)

4. WHY THIS RESULT WAS GIVEN

- Insecure Transport Protocol:** URL uses plaintext HTTP instead of encrypted HTTPS, leaving traffic open to interception.
- Credential Harvesting Tokens:** Detected 3 targeted security/authentication keyword(s) ('signin, verify, account') in the URL path.
- Brand Typosquatting / Combosquatting:** Domain incorporates deceptive hyphenated tokens to impersonate authentic brand assets.
- High-Risk Top-Level Domain:** Registered under a TLD associated with elevated abuse and disposable phishing campaigns.
- Elevated Character Randomness (Entropy: 4.74):** String exhibits algorithmic randomness typical of automated phishing kits.

5. EXPLAINABLE AI SUMMARY (SHAP & LIME)

Explainable AI mathematically attributes how individual URL characteristics pushed the machine learning model toward **Phishing (+Risk)** or **Legitimate (-Risk)**:

Ran k	Influential Feature	Observed Value	Risk Impact Direction	Diagnostic Influence Summary
#1	TLD Risk Factor	0.8	+Risk (Pushed Phishing)	Contributed +0.244 weight to prediction
#2	Subdomain Count	1	-Risk (Pushed Safe)	Contributed -0.164 weight to prediction
#3	Suspicious Keywords Count	3	+Risk (Pushed Phishing)	Contributed +0.089 weight to prediction

#4	HTTPS Protocol	0	+Risk (Pushed Phishing)	Contributed +0.052 weight to prediction
#5	Path Length	24	-Risk (Pushed Safe)	Contributed -0.043 weight to prediction

6. SECURITY RECOMMENDATIONS

■ HIGH-RISK INCIDENT REMEDIATION DIRECTIVES

- **Do not enter username or password:** This page is designed to harvest credentials.
- **Do not provide personal or financial information:** Avoid submitting payment details, SSNs, or identity data.
- **Do not download files from the URL:** Executables or documents served from this host may contain malware.
- **Verify the domain using a trusted source:** Navigate directly to official services via verified bookmarks or search engines.
- **Close and report the suspicious page:** Terminate active browser tabs and report this URL to your IT/SOC security team.

7. SHORT SECURITY SUMMARY

Target URL:	http://paypal-security-update.account-verify.xyz/signin.php?token=928103	Risk Level:	CRITICAL
Final Verdict:	PHISHING	Confidence:	85.0% (Phishing: 85.0%)
Main Reason:	Detected malicious indicators: 5 structural/lexical risk factors (Phishing Probability: 85.0%).	Recommended Action:	BLOCK & DO NOT PROCEED. Do not submit credentials or download files.